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MIL-H-82000C(YD)
19 March 1990
SUPERSEDING
MIL-H-82000B(YD)
20 April 1984

MILITARY SPECIFICATION

HEATERS, HOT OIL, 1.5, 3, AND 5 MILLION BTU/H OUTPUT, SKID MOUNTED

This specification is approved for use by the Naval Facilities Engineering Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers high temperature, self-contained, fully automatic, circulation hot oil heaters with 1,500,000, 3,000,000, and 5,000,000 British thermal units per hour (Btu/h) minimum output. The heaters will be skid mounted, light fuel oil-fired, and electric-driven, for use as heat producing and transfer agents.

1.2 Classification. The heaters will be one of the following sizes, as specified (see 6.2).

Size 1.5 - 1,500,000 Btu/h minimum output.
Size 3 - 3,000,000 Btu/h minimum output.
Size 5 - 5,000,000 Btu/h minimum output.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer (Code 156), Naval Construction Battalion Center, Port Hueneme, CA 93043-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 3895

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2.1.1 Specifications and standard. The following specifications and standard, form a part of this document to the extent specified herein. Unless otherwise specified the issues of these documents are those listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

VV-F-800 - Fuel Oil, Diesel.

MILITARY

MIL-F-3296 - Forges, Furnaces and Ovens (Exclusive of Space Heating and Cooking), Packaging of.
MIL-T-5624 - Turbine Fuel, Aviation, Grades JP-4 and JP-5.
MIL-L-22851 - Lubricating Oil, Aircraft Piston Engine, (Ashless Dispersant).

STANDARD

MILITARY

MIL-STD-209 - Slinging Eyes and Attachments for Lifting and Tying Down Military Equipment.

(Unless otherwise indicated, copies of federal and military specifications and standard are available from the Military Specifications and Standards. Bldg 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094)

2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents which are current on the date of the solicitation (see 6.2).

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME Boiler and Pressure Vessel Code.
Section VIII, Division 1 - Pressure Vessels.
ASME B31.1 - Power Piping.

(Application for copies should be addressed to the American Society of Mechanical Engineers, United Engineering Center, 345 East 47th Street, New York, NY 10017.)

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NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

NEMA MG 1 - Motors and Generators. (DoD adopted)
NEMA ICS 6 - Enclosures for Industrial Controls and Systems. (DoD adopted)

(Application for copies should be addressed to the National Electrical Manufacturers Association, 2101 L Street, N.W., Washington, DC 20037.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 70 - National Electrical Code.

(Application for copies should be address to the National Fire Protection Association, Batterymarch Park, Quincy, MA 02269.)

SOCIETY OF AUTOMOTIVE ENGINEERS, INC. (SAE)

SAE J534 - Lubrication Fittings. (DoD adopted) .

(Application for copies should be addressed to the Society of Automotive Engineers, Inc., 400 Commonwealth Drive, Warrendale, PA 15096.)

UNDERWRITERS LABORATORIES (UL), INC. (UL)

UL 674 - Standard for Electric Motors and Generators for use in Hazardous Locations, Class I, Groups C and D, Class II, Groups E, F, and G.

(Application for copies should be addressed to the Underwriters Laboratories, Inc., 333 Pfingsten Road, Northbrook, IL 60062.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this specification shall take precedence. Nothing in this document, however, supersedes applicable laws and regulations unless specific exemption has been obtained.

3. REQUIREMENTS

3.1 Description. The heater shall consist essentially of an insulated heating chamber, a light fuel oil burner system, a separate expansion tank or internal provision for expansion, circulating system, controls, safety devices, and all necessary accessories to make a complete self-contained unit, ready for use after service connections have been made. The heater and all controls or components shall perform and operate efficiently as a unit, at a starting ambient temperature of +35 degrees Fahrenheit (^{o}F) and continue to operate in the event of a temperature drop to -25°F. In addition, the fuel consumption of

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the heater, at rated output, operating at low ambient temperature, using the specified light fuel oil as a heat producing agent, shall not exceed 16 gallons per hour (gph) for size 1.5 heater, 32 gph for size 3 heater, and 53 gph for size 5 heater. Light fuel oil herein specified for heat generating purpose shall be any of the following or combination thereof: DF-A, DF-1, and DF-2 diesel oil conforming to VV-F-800, or JP-4 and JP-5 jet fuel conforming to MIL-T-5624.

3.2 First article. When specified, the contractor shall furnish one complete heater for first article inspection and approval (see 4.2.1, 6.2, and 6.3).

3.3 Materials. Materials used shall be free from defects which would adversely affect the performance or maintainability of individual components or of the overall assembly. Materials not specified herein shall be of the same quality used for the intended purpose in commercial practice. Unless otherwise specified herein, all equipment, material, and articles incorporated in the work covered by this specification are to be new and fabricated using materials produced from recovered materials to the maximum extent possible without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. Unless otherwise specified, none of the above shall be interpreted to mean that the use of used or rebuilt products are allowed under this specification. Unless otherwise specified.

3.3.1 Transfer oil. Transfer oil shall not be furnished with the heater. Transfer oil used in the heat transfer system shall be of the high quality petroleum stock, the properties of which shall conform to MIL-L-22851, type III.

3.3.2 Piping and tubing. All internal piping shall be seamless steel tubing or piping in accordance with ANSI B31.1 for 150 pound-force per square inch (psi), 450°F service

3.3.3 Valves and pipe fittings. All valves, flanges, fittings, unions, and nipples shall be of steel in accordance with ANSI B31.1 for 150 psi, 450°F service. Fittings shall be screwed American National taper pipe threads (NPT), or flanged connections and, when made up, shall be capable of withstanding a 150 psi at a temperature of 450°F.

3.4 Safety. With the exception of transmission lines, overflow lines, cold oil seal tank (if furnished), and the combustion head, all exposed parts that are subject to high operating temperatures or that are energized electrically shall be insulated, enclosed, or guarded. All moving parts that are of such nature or so located as to be a hazard to operating or maintenance personnel shall be enclosed or guarded. Protective devices shall not impair the operating functions of the heater. A ground rod shall be provided to ground the heater during operation.

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3.5 Ease of maintenance. The unit shall permit accessibility to all parts for maintenance and repair utilizing common mechanic handtools and the minimum of special service tools, furnished by manufacturer, associated with items of this nature. Locking devices and controls with threaded fasteners, that can be overtorqued by operators, shall not be used. Covers or plates that must be removed for component adjustment or for component removal shall be equipped with quick-disconnect fasteners.

3.6 Performance. The heater, burner, and main circulating pump shall be capable of performing as follows:

a. The maximum heater output capacity shall be not less than 1,500,000 Btu/h at a transfer oil temperature of 430°F for size 1.5, 3,000,000 Btu/h for size 3, and not less than 5,000,000 Btu/h for size 5, when charged with the type of transfer oil specified in 3.3.1.
b. The maximum fuel oil consumption of the burner shall be 16 gph for size 1.5, 32 gph for size 3, and 52 gph for size 5.
c. The circulating pump shall have a minimum capacity, as an oil temperature not greater than 430°F and a pump discharge pressure of not less than 50 pound-force per square inch gage (psig), of not less than 130 gallons per minute (gpm) of transfer oil for size 1.5 heater, 260 gpm for size 3 heater, and 350 gpm for size 5 heater.
d. The heater shall be provided with a fully automatic fire operating sequence system for safeguarding against flame failure as specified in 3.10.1.
e. The heater shall be equipped with automatic operating safety controls as specified in 3.10.2.

3.7 Power.

3.7.1 Power supply. All electric power necessary to operate the heater shall be supplied from an outside source and be nominal 480 volts (V), 3 phase, 60 Hertz (Hz), alternating current (ac), supplied through a weathertight main power panel with main breaker and circuit breaker protected circuits. Where required, receptacle shall be weathertight and mounted on the most accessible space of the unit and away from the heated area.

3.7.2 Motors. All motors shall be 230/460 V, 3 phase, 60 Hz, ac, induction, totally enclosed, air cooled conforming to NEMA MG 1, unless otherwise specified (see 6.2).

3.7.2.1 Motor starters. Motors shall be provided with magnetic starter of 460 V rating. Starter enclosure shall be type 3S in accordance with NEMA ICS 6.

3.7.2.2 Wiring. All wiring shall be conduit installed and conform to the wiring practices and design of NFPA 70.

3.7.2.3 Hazardous locations. When specified for use in hazardous locations, class and types per NFPA 70 (see 6.2), the motor shall conform to UL 674, controller to applicable NEMA ICS 6 hazardous class, and wiring to NFPA 70 hazardous locations.

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3.7.3 Drives.

3.7.3.1 V-belt drives. V-belts shall be in accordance with MIL-B-11040, suitably modified for use of fractional horsepower V-belt drives and shall be adjustable.

3.7.3.2 Reducing-gear drives. Reducing-gear drives shall be totally enclosed and shall operate in an oil bath.

3.8 Heating chamber. The heating chamber shall be light fuel oil-fired, shall be either the fire-tube, liquid-tube, or complete jacketed type, and shall be of all steel construction with the quality of welding and fabrication to conform to the intent of the ASME Boiler and Pressure Vessel Code, section IX, for a working pressure of 100 psig. The heater shall maintain not less than 130 gpm for size 1.5 heater, 260 gpm for size 3 heater, and 350 gpm for size 5 heater, of transfer oil at a maximum outlet temperature of 430°F when the inlet temperature of transfer oil is not higher than 380°F. The heating chamber shall be provided with a burner combustion ring of refractory material having a pyrometric cone equivalent of not less than 32. The transfer oil at the inlet of the heating chamber shall have a pressure not greater than 15 psig.

3.9 Insulation. The heating chamber shall be insulated with the exception of the combustion head, oil transmission line, cold oil seal tank, and overflow lines. The insulation shall be covered with a sheet steel outer shell not less than 0.05989-inch thick. The insulation shall maintain the temperature of the outer shell at not more than 125°F plus the ambient temperature after the heater has been operated at rated output capacity for not less than 4 hours. If the temperature of the outer shell is greater than 850°F plus the ambient temperature after the heater has been operated at rated output capacity for not less than 4 hours, the outer shell shall be shielded or guarded as specified in 3.4.

3.10 Burner. The burner shall be a light fuel oil burner equipped with pilot and shall be of the air-atomizing or mechanical-atomizing type, capable of burning any type of light fuel oil as specified in 3.1 or combinations thereof. The burner shall be equipped with direct electrical ignition of the pilot and shall be sequenced for fully automatic operation. The combustion controls shall be of the on-off type, three-position high-low-off type, or of the modulating type. The burner turndown ratio of the three-position type shall be not less than 3 to 1. The low firing rate, where applicable, shall be approximately 33-1/3 percent of the maximum firing rate. The flame-failure reaction time shall be not more than 4 seconds. The burner unit shall be provided with electric-driven pumps, fans, blowers, and other accessories necessary for operation. The fuel line shall include a duplex-type strainer to permit continuous operation while changing one strainer element.

3.10.1 Fire operating sequence. The fire operating sequence shall be as follows:

a. Burner initial start by means of a manual starting switch.
b. Fan and timer start for four air changes to purge the heater. The total purge time shall be not less than 15 seconds.

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c. After completion of purge, when all fuel safety and heater interlocks are closed, the combustion controls shall actuate the igniter and open the main fuel safety shutoff valve.
d. When the main fuel valve opens, the trial for ignition period shall begin. When the flame-sensing device does not sense the pilot or low flame within 15 seconds after the main fuel valve opens, the device shall lock the burner out to prevent automatic recycling, unless the time of ignition trial periods are of 3 seconds or less, in which case automatic recycling shall be terminated after not more than 13 trials. When this occurs, a manual resetting of the combustion controls shall be required.
e. After the low flame is detected, the combustion controls shall adjust to heat demand. On high-low type or modulating type, the burner shall go to high-fire position from low-fire position.
f. The burner shall go from high-fire position to low-fire position upon satisfying the heat demand and shall return to high-fire position upon a new call for heat. The modulating type burner shall adjust to heat demand throughout the entire modulating range. The on-off type shall be on or off in accordance with heat demand and shall automatically recycle to these positions.
g. Overheating, fuel safety interlock opening, or oil level falling below minimum shall shut down the burner automatically, and when the condition has been sufficiently corrected, the burner shall be restarted.

3.10.2 Controls and safety devices. Controls and safety devices shall include the following:

a. Indicating temperature regulator of the knob type that can be set manually for the desired transfer oil temperature. The regulator shall be adjustable from 100°F with a suitable clamp affixed to the scale up to 450°F painted red. A thermometer registering up to 500°F shall be provided to indicate the actual temperature for comparison with the setting.
b. High-temperature limit control that shall cut off the burner in the event the regulator fails. This control shall be adjustable from 100°F to 450°F and shall be set by the manufacturer at 450°F.
c. Low oil-level cutoff control that shall cut off the burner and main circulating pump when the oil level in the expansion tank falls below 25 percent of capacity level.
d. Flame-sensing device of the ultraviolet-detector, photo-electric-cell, or frequency-response-cell type. The main oil flame shall be sensed as long as it is present.
e. The flame-failure control shall be of the electronic type and shall have a control reaction time of not more than 4 seconds.
f. A control to cut off the burner in event of failure of the main circulating pump.
g. All necessary interlocks and accessories to insure sequence of operation as specified in 3.11 and to insure operation of all controls and safety devices.

3.11 Transfer oil pumping and piping system.

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3.11.1 Circulating pump. The circulating pump shall be either:

a. Positive displacement or centrifugal.
b. Vertical or horizontal.

Horizontal or vertical type pumps shall be provided with antifriction bearings or plain bushing type bearings. All bearings shall be either self-lubricated or greaseable and lubricated from an external source. Lubrication of bearings with transfer oil shall be prohibited. Pumps shall be operable at a temperature of 450°F equipped with high temperature leakproof packing or other means of mechanical seal.

3.11.2 Piping system. The piping system shall provide all necessary valves and fittings to interconnect the heating chamber, pump or pumps, and the expansion tank and shall terminate with an inlet and outlet service connection.

Service connections shall be 2-1/2 inch iron pipe size NPT for size 1.5 heater, and 3-inch iron pipe size NPT for size 3 and 5 heaters, and shall be equipped with caps and cap retaining chains, or terminal valves.

3.11.2.1 Strainer. A single strainer of the full flow "Y" type, with valves on each side, shall be installed into the return line. The strainer body, with moisture or sediment separation provision, shall be of steel and the strainer element shall be steel screen of corrosion-resistant quality.

3.11.2.2 Pump location. The circulating pump shall be located so as to assure continuous flow through the heating system regardless of flow, and so that there shall be a positive head of transfer oil on the suction side at all times.

3.12 Expansion tank. A steel expansion tank, or an internal space for expansion, shall be provided. The tank, or integral provision, shall consist essentially of a space to compensate for the expansion of transfer oil being heated in the heating chamber and that being circulated to the external user system, a removable filler cap, a means for filling, a means for visually determining the oil level, a drain, a means of venting all necessary controls, and all other accessories necessary for rated heater output operation. The expansion tank shall be isolated from the heating chamber and shall be insulated. The expansion tank, or space for expansion, shall be located at an elevation above all pumps. The arrangement shall be such that when the unit is operating, the oil level in the facility for expansion merely rises due to expansion of oil being heated, and falls due to cessation of heating. The minimum capacity of the facility for expansion shall be such that, when the transfer oil is at the normal operating level in the heater and in the facility for expansion at 430°F, there shall be a net reserve for expansion of oil in the external user system of not less than 65 gallons for size 1.5 heaters, 120 gallons for the size 3 heater, and not less than 165 gallons for the size 5 heater.

3.13 Frame. The frame shall be fabricated of structural steel shapes, suitably gusseted and braced, so that there will be no sagging when supporting a full load or distortion of the unit in rough handling during transportation.

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3.14 Interchangeability. All units of the same classification furnished with similar options under a specific contract shall be identical to the extent necessary to insure interchangeability of component parts, assemblies, accessories, and spare parts.

3.15 Lubrication. Unless otherwise specified (see 6.2), means for lubrication shall be in accordance with the manufacturer's standard practice. The lubricating points shall be easily visible and accessible. All parts requiring lubrication shall be lubricated as specified in section 5. Hydraulic lubrication fittings shall be in accordance with SAE J534. Where use of high pressure lubricating equipment, 1,000 psi or higher, will damage grease seals or other parts, a suitable warning shall be affixed to the equipment in a conspicuous location.

3.16 Cleaning, treatment, and painting. Surfaces normally painted in good commercial practice shall be cleaned, treated, and painted as specified herein. The color of the finish coat shall be as specified (see 6.2). Surfaces to be painted shall be cleaned and dried to insure that they are free from contaminants such as oil, grease, welding slag and spatter, loose mill scale, water, dirt, corrosion product, or any other contaminating substances. As soon as practicable after cleaning, and before any corrosion product or other contamination can result, the surfaces shall be prepared or treated to insure the adhesion of the coating system. The painting shall consist of at least one coat of primer and one finish coat. The primer shall be applied to a clean, dry surface as soon as practicable after cleaning, treating, and Painting shall be with manufacturer's current materials according to manufacturer's current processes and the total dry film thickness shall be not less than 2.5 mils over the entire surface. The paint shall be free from runs, sags, orange peel, or other defects.

3.17 Fungus resistance. Electrical components and circuit elements, including terminal and circuit connections, shall be coated in accordance with the manufacturers standard practice.

3.18 Lifting and tiedown attachments. When specified (see 6.2), the heater shall be equipped with lifting and tying down attachments. Lifting and tiedown attachments shall conform to type II or type III of MIL-STD-209. A nonferrous transportation plate shall be provided and mechanically attached to the heater. Transportation plates shall be inscribed with a diagram showing the 10 lifting attachments and lifting slings, the capacity of each attachment, and the required length and size of each sling cable. A silhouette of the item furnished showing the center of gravity shall be provided on the transportation plate. Tying down attachments may be identified by stenciling or other suitable marking. Tiedown marking shall clearly indicate that the attachments are intended for the tie down of the heater on the carrier when shipped.

3.19 Toolbox. The heater shall be provided with a compartment storage space for tools and technical publications, or a toolbox normally furnished on commercial heaters.

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3.20 Instruction plates. The heater shall be equipped with instruction plates suitably located, describing any special or important procedures to be followed in operating and servicing the equipment. Plates shall be of a material which will last and remain legible for the life of the equipment, and shall be securely affixed thereto.

3.21 Identification plate. An identification plate will be furnished by the contracting officer for each heater. The contractor shall stamp all necessary data in the blank spaces of the plate provided for that purpose, and securely affix a plate to each heater in a conspicuous place. The applicable nomenclature contained in the contract item description shall be placed in the top blank.

3.22 Workmanship.

3.22.1 Steel fabrication. The steel used in fabrication shall be free from kinks, sharp bends, and other conditions which would be deleterious to the finished product. Manufacturing processes shall not reduce the strength of the steel to a value less than intended by the design. Manufacturing processes shall be done neatly and accurately. All bends shall be made by controlled means to insure uniformity of size and shape.

3.22.2 Bolted connections. Bolt holes shall be accurately punched or drilled and shall have the burrs removed. Washers or lockwashers shall be provided in accordance with good commercial practice, and all bolts, nuts, and screws shall be tight.

3.22.3 Riveted connections. Rivet holes shall be accurately punched or drilled and shall have the burrs removed. Rivets shall be driven with pressure tools and shall completely fill the holes. Rivet heads, when not countersunk or flattened, shall be of approved shape and of uniform size for the same diameter of rivet. Rivet heads shall be full, neatly made, concentric with the rivet holes, and in full contact with the surface of the member.

3.22.4 Welding. Welding procedures shall be in accordance with a nationally recognized welding code. The surface of parts to be welded shall be free from rust, scale, paint, grease, or other foreign matter. Welds shall be of sufficient size and shape to develop the full strength of the parts connected by the welds. Welds shall transmit stress without permanent deformation or failure when the parts connected by the weld are subjected to proof and service loadings.

3.22.5 Castings. All castings shall be sound and free from patching, misplaced coring, warping, or any other defect which reduces the castings ability to perform its intended function.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor

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may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 Component and material inspection. Components and materials shall be inspected in accordance with all the requirements specified herein and in applicable referenced documents.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2.1).
b. Quality conformance inspection (see 4.2.2).

4.2.1 First article inspection. The first article inspection shall be performed on the heater when a first article is required (see 3.2, 6.2 and 6.3). This inspection shall include the examination of 4.3, the tests of 4.4, and, when specified, the preproduction pack inspection of 4.5 (see 4.5 and 6.2). The first article may be either a first production item or a standard production item from the supplier's current inventory provided the item meets the requirements of the specification and is representative of the design, construction, and manufacturing technique applicable to the remaining items to be furnished under the contract.

4.2.2 Quality conformance inspection. The quality conformance inspection shall include the examination of 4.3, the test of 4.4.2(g), the packaging inspection of 4.5. The inspection shall be performed on the samples selected in accordance with 4.3.

4.3 Examination. Each hot oil heater shall be examined for compliance with the requirements specified in section 3 of this specification. Any redesign or modification of the contractor's standard product to comply with specified requirements, or any necessary redesign or modification following failure to meet specified requirements shall receive particular attention for adequacy and suitability. This element of inspection shall encompass all visual examinations and dimensional measurements. Noncompliance with any specified requirements or presence of one or more defects preventing or lessening maximum efficiency shall constitute cause for rejection.

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4.4 Tests. The first article, or any heater being selected, shall be tested in accordance with 4.4.1 through 4.4.4. Failure of the heater to pass any test shall be the basis for rejection.

4.4.1 Test conditions. Prior to examination or test, the heater shall be lubricated. Oils shall be those designated for use in the ambient temperatures at which the test is being performed. When services and adjustments are performed, the heater shall be checked for ease of maintenance.

4.4.2 Performance test. The heater shall be subjected to the following tests:

a. Heater temperature and flow. The heater shall be fully charged with transfer oil (see 3.3.1), and operated until the discharge flow is stabilized at 130 gpm (+5 gpm -0 gpm) for size 1.5 heater at a maximum temperature of 430°F; 260 gpm (+5 gpm -0 gpm) for size 3 heater at 430°F; and 350 gpm (+5 gpm -0 gpm) for size 5 heater at 430°F. The oil shall flow through a closed circuit to the return port on the heater at which point the temperature of the transfer oil shall be not higher than 380°F, and the pressure shall be not greater than 15 psig.
b. Burner. The heater shall be operated for not less than 1 hour under conditions specified in 4.4.2a, and burner light fuel consumption shall be recorded to determine conformance to 3.1.
c. Pump. The circulating pump shall be operated for 1 hour while pumping transfer oil at a temperature of not more than 430°F and at a discharge pressure of 50 psig to determine conformance to 3.6c.
d. Heating chamber insulation. The heater shall be operated at rated capacity for not less than 4 hours, at which time the temperature of the outer shell shall be measured at various locations to determine conformance to 3.9.
e. Control and safety devices. Each control and safety device shall be subjected to functional operation for not less than 10 times each. Control and safety devices shall function as specified in 3.10.2.
f. Endurance. The heaters shall be operated for not less than 8 hours at minimum capacity of not less than 130 gpm and not less than 430°F for 1.5 size heater; 260 gpm for size 3 heater; and 350 gpm for size 5 heater. The heaters may be operated continuously or in increments of not less than 6 hours. Any evidence of leakage, breakage, permanent distortion, or malfunction shall constitute failure of this test.
g. Cycling. The heater shall be started and restarted for 15 times within a period of 1 hour. Failure of 3 operating sequences as specified in 3.10.1, or any evidence of leakage or malfunction, shall constitute failure of this test.

4.4.3 Overloading safety factor tests. The following overloading safety factor tests shall be conducted. Any structural, mechanical, and control failures or other malfunctions due to these subsequent critical tests shall be considered as the basis for rejection.

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a. Prior to the heater being charged with transfer oil, all thermometers and temperature controls designed for the maximum operating temperatures of 450°F to 500°F shall be removed and replaced with interim sets having the capacity to withstand higher temperatures up to 625°F. The heater shall then be fully charged with the transfer oil and tested with continuous firing at full firing rate to obtain hot oil at 600°F for 24 consecutive hours with discharge flow stabilized at 130 gpm (+5 gpm -0 gpm) for size 1.5 heater; 260 gpm (+5 gpm -0 gpm) for size 3 heater; and at 350 gpm (+5 gpm -0 gpm) for size 5 heater; and a temperature differential, of the oil leaving the heater and entering the heater, of not less than 40°F.
b. The heater shall then be tested for 25 continuous, repeated, and recorded complete firing cycles from minimum to maximum firing rate. Each complete cycle shall not exceed 5 minutes beyond minimum possible actuation time for any particular burner control system.

4.4.4 Lifting and tying down attachment tests. When required, the lifting and tying down attachments shall be tested to conform to 3.18.

4.5 Packaging inspection. The inspection of the preservation, packing, and marking shall be in accordance with the requirements of section 4 of MIL-F-3296. The inspection shall consist of the quality conformance inspection; and, when specified (see 6.2), a preproduction pack shall be furnished for examination and test within the time frame required (see 6.2).

5. PACKAGING

5.1 Preservation, packing, and marking. Preservation, packing, and marking shall be in accordance with the requirements of MIL-F-3296 with the level of preservation and the level of packing as specified (see 6.2).

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The heaters are intended to heat and circulate transfer oil, as a heat transfer medium up to a temperature of 430 F for use with asphalt plants, asphalt melters, or similar heating applications.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Number, title, and date of this specification.
b. Size of heater required (see 1.2).
c. Issue of DODISS to be cited in the solicitation and if required, the specific issue of individual documents cited (see 2.1.1 and 2.2).
d. When a first article is required for inspection and approval (see 3.2, 4.2.1, and 6.3).
e. When electric motor requirement is other than as specified (see 3.7.2).
f. When used in hazardous locations, class, and type per NFPA 70 required (see 3.7.2.3).

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g. When lubrication shall not be in accordance with manufacturer's standard practice (see 3.15).
h. Color of finish coat required (see 3.16).
i. When lifting and tying down attachments are required (see 3.18).
j. When a preproduction pack inspection is required and the time frame required for submission (see 4.2.1 and 4.5).
l. Level of preservation and level of packing required (see 5.1).

6.3 First article. When a first article inspection is required (see 6.2), the item will be tested and should be a first production item or it may be a standard production item from the contractor's current inventory as specified in 4.2.1. The first article should consist of one unit. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, test, and approval of the first article.

6.4 Subject term (key word) listing.

Burner
Circulating pump
Expansion tank
Heating chamber
Light fuel oil
Strainer
Temperature regulator
V-Belt drive

6.5 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodian:
Navy - YD

Preparing Activity:
Navy - YD

(Project 3895-N278)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
3. DOCUMENT TITLE		

4. NATURE OF CHANGE (*Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.*)

5. REASON FOR RECOMMENDATION

6. SUBMITTER		
a. NAME (*Last, First, Middle Initial*)	b. ORGANIZATION	
c. ADDRESS (*Include Zip Code*)	d. TELEPHONE (*Include Area Code*) (1) Commercial (2) AUTOVON (*If applicable*)	e. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY	
a. NAME	b. TELEPHONE (*Include Area Code*) (1) Commercial (2) AUTOVON
c. ADDRESS (*Include Zip Code*)	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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